

Multi-Team Scaling Scenarios

- **Scenario 1 - 2 Teams (12 employees in Total)**
 - Team A (10 employees)
 - Team B (5 Employees)
 - 3 members shared between teams

- **Scenario 2 - 4 Teams (24 employees in Total)**
 - Team A (7 employees)
 - Team B (7 employees)
 - Team C (7 employees)
 - Team D (7 employees)
 - 4 members have one of the teams as a 2nd preference

- **Scenario 3 - 8 Teams (48 employees in Total)**
 - Team A (7 employees)
 - Team B (7 employees)
 - Team C (7 employees)
 - Team D (7 employees)
 - Team E (7 employees)
 - Team F (7 employees)
 - Team G (7 employees)
 - Team H (7 employees)
 - 8 members have one of the teams as a 2nd preference

- **Scenario 4 - 16 Teams (96 employees in Total)**
 - Team A (7 employees)
 - Team B (7 employees)
 - Team C (7 employees)
 - Team D (7 employees)
 - Team E (7 employees)
 - Team F (7 employees)
 - Team G (7 employees)
 - Team H (7 employees)
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 - Team P (7 employees)
 - 16 members have one of the teams as a 2nd preference

4 different scenarios for vacation generation:

- **Scenario 1 — Sequential Rotation (Base method)**
 - Employees take 30 consecutive vacation days, distributed sequentially throughout the year.
 - For example the first employee starts a 30 day block on January 1st and the following employee begins after the previous one ends (In this case on January 31st). If one of the employees have their vacation at the end of the year it will continue at the beginning of the year.
 - This guarantees that vacations days are evenly distributed
- **Scenario 2 — Random Two-Block Distribution**
 - Employees are assigned vacations on two random blocks of 15 days across the calendar year
 - Guaranteed that these two blocks won't overlap with one another (to make sure the total of vacation days is always 30)
- **Scenario 3 — Weekend Focused Short Breaks**
 - Considering employees can work every year during any day, this scenario was created to mimic a more real human vacation scenario
 - Employees take multiple 4–6-day breaks, distributed throughout the year and biased toward weekends
 - Each employee's set of breaks totals **30 days**, spread over ~6–8 mini-vacations spaced about 3 to 5 weeks apart.
- **Scenario 4 — Summer-Focused Staggered Vacations**
 - All employees' vacations are restricted to the **summer period (June 1 – August 31)**.
 - Each employee takes **one continuous 30-day vacation**, with start dates evenly distributed within this 3-month window.
 - Used to represent seasonal vacation spikes

Minimum Templates Generation

To simulate the operational staffing requirements corresponding to each multi-team scenario,

a **Minimums Template Generator** was developed.

This generator scales the daily *minimum* and *ideal* staffing levels from the **base configuration scenario** (2 teams, 15 employees).

Scaling Methodology

For each target scenario (4, 8, and 16 teams),

The script performs proportional scaling and controlled variation as follows:

1. Per-Team Scaling Factor

The number of required employees per team is scaled by the ratio of target to base headcount:

Scale Factor

$$= \frac{\text{Employees per Team (Target) (excluding shared)}}{\text{Employees per Team (Base)}}$$

For example, if each team in the base scenario has 10 employees and the target scenario has 7 per team, all the daily minimums are multiplied by **0.7**. After multiplying they are then transformed into integers (using banker's rounding [e.g 2.5→2, 3.5→4])

2. Controlled Random Variation (± 1)

To avoid the templates to be exactly the same per team, a small random ± 1 perturbation was applied to a small fraction of days ($\approx 4-5\%$). These perturbations are balanced meaning that the number of +1 and -1 changes are equal to guarantee that the scaling remains constant.

3. Ideal Staffing

$$Ideal(d, t, e) = Minimum(d, t, e) + 1$$

To guarantee that if one of the members of a team has to miss a work day due to an unexpected situation the minimums are still covered.

Total Complete Scenarios -> 4 Multi-Team Scaling * Scenarios 4 Vacation Scenarios =

16 Complete Scenarios

Considering we are testing ILP and CSP, this makes up for 32 algorithmic results (16*2)

Results

ILP (Maximum time - 45 minutes)

Weights used - 100 minimums, 1 ideals

	Time Taken	Minimuns not accomplished	Ideals not accomplished	% of minimums not filled	% of ideals not filled
2Teams_Vacation1	4.39 sec	16	126	0.77%	5.04%
2Teams_Vacation2	4.60 sec	16	127	0.77%	5.08%
2Teams_Vacation3	3.21 sec	16	127	0.77%	5.08%
2Teams_Vacation4	4.10 sec	48	192	2.3%	7.67%
4Teams_Vacation1	16.39 sec	0	488	0%	8.35%
4Teams_Vacation2	19.53 sec	0	509	0%	8.71%
4Teams_Vacation3	20.55 sec	0	492	0%	8.42%
4Teams_Vacation4	15.25 sec	0	693	0%	11.86%
8Teams_Vacation1	56.92 sec	0	976	0%	8.36%
8Teams_Vacation2	1 min 34 sec	0	987	0%	8.45%
8Teams_Vacation3	1 min 9 sec	0	976	0%	8.36%
8Teams_Vacation4	1 min 16 sec	0	1374	0%	11.76%
16Teams_Vacation1	4 min 53 sec	0	1952	0%	8.35%
16Teams_Vacation2	4 min 53 sec	3	2008	0.03%	8.60%
16Teams_Vacation3	4 min 45 sec	0	1960	0%	8.39%
16Teams_Vacation4	5 min 49 sec	67	2824	0.57%	12.09%

CSP (Maximum time - 45 minutes)

Weights used - 100 minimums, 1 ideals

	Time Taken	Minimuns not accomplished	Ideals not accomplished	% of minimums not filled	% of ideals not filled
2Teams_Vacation1	23.588 sec	16	127	0.77%	5.08%
2Teams_Vacation2	32.38 sec	16	128	0.77%	5.12%
2Teams_Vacation3	48.94 sec	16	128	0.77%	5.12%
2Teams_Vacation4	58.22 sec	48	192	2.3%	7.67%
4Teams_Vacation1	4 min 54 sec	0	488	0%	8.35%
4Teams_Vacation2	9 min 45 sec	0	509	0%	8.71%
4Teams_Vacation3	9 min 15 sec	0	492	0%	8.42%
4Teams_Vacation4	45 min 1 sec	0	695	0%	11.87%
8Teams_Vacation1	16 min 3 sec	0	976	0%	8.36%
8Teams_Vacation2	45 min 3 sec	0	991	0%	8.48%
8Teams_Vacation3	16 min 4 sec	0	976	0%	8.36%
8Teams_Vacation4	45 min 2 sec	0	1385	0%	11.76%
16Teams_Vacation1	45 min 5 sec	0	1956	0%	8.37%
16Teams_Vacation2	45 min 8 sec	3	2024	0.03%	8.66%
16Teams_Vacation3	45 min 10 sec	0	1976	0%	8.46%
16Teams_Vacation4	45 min 8 sec	71	2852	0.61%	12.20%

GA

Weights used - 100 minimums, 1 ideals

The genetic algorithm (GA) was evaluated across all multi-team scaling scenarios, with ten independent runs conducted for each scenario. The results represent the average of the 10 runs; in contrast, ILP and CSP are single deterministic runs. The configuration was as follows: population size = 200; NBTS crossover; demand-guided mutation (gene mutation probability = 0.003); tournament size = 7; early stopping with patience = 100 generations; and minimum delta = 1. 'Avg. number of generations to stop' reflects the average generation at which no meaningful improvement was detected.

	Time Taken	Minimums not accomplished	Ideals not accomplished	% of minimums not filled	% of ideals not filled	Avg. number of generations to stop
2Teams_Vacation1	2 min 7 sec	23	129.6	1.10%	5.18%	549.9
2Teams_Vacation2	2 min 55 sec	21.8	139.8	1.05%	5.59%	663.2
2Teams_Vacation3	2 min 17 sec	22.8	131.4	1.09%	5.25%	580.7
2Teams_Vacation4	2 min 9 sec	60.7	201.4	2.91%	8.05%	546.9
4Teams_Vacation1	6 min 22 sec	0.3	558.3	0.01%	9.56%	866.4
4Teams_Vacation2	min sec					
4Teams_Vacation3	min sec					
4Teams_Vacation4	min sec					
8Teams_Vacation1	12 min 48 sec	0.5	1171.0	0.01%	10.03%	874.1
8Teams_Vacation2	min sec					
8Teams_Vacation3	min sec					
8Teams_Vacation4	min sec					
16Teams_Vacation1	21 min 29 sec	7.2	2396.9	0.06%	10.26%	752.0
16Teams_Vacation2	min sec					
16Teams_Vacation3	min sec					
16Teams_Vacation4	min sec					
32Teams_Vacation1	63 min 14 sec	21.6	4659.3	0.09%	9.98%	796.1
32Teams_Vacation2						

32Teams_Vacation3						
32Teams_Vacation4						

Vacation Scenarios 2, 3 and 4 to be run on reference machine for publication.

GA + Local Search

Weights used - 100 minimums, 1 ideals

After the GA has finished, a local post-processing search is applied to the best solution found. This phase alternates between two operators: a day-swap operator, which reassigns surplus shifts to cover deficit slots, and a cyclic exchange operator, which rotates assignments across employees, until no further improvement is achieved. The local search only targets ideal coverage, as minimum coverage is already handled by the GA's repair mechanism. The time shown is the post-processing time only; 'Total time' includes both the GA and the local search. Despite adding only 1–5 seconds, the local search consistently reduces ideals not accomplished by 3–8%.

	Time Taken	Ideals not accomplished	% of ideals not filled	Total time
2Teams_Vacation1	1.83 sec	128.8	5.15%	2 min 9 sec
2Teams_Vacation2	1.84 sec	139.1	5.56%	2 min 57 sec
2Teams_Vacation3	1.46 sec	131.2	5.24%	2 min 18 sec
2Teams_Vacation4	2.68 sec	200.8	8.03%	2 min 12 sec
4Teams_Vacation1	0.56 sec	550.5	9.43%	6 min 23 sec
4Teams_Vacation2	sec			
4Teams_Vacation3	sec			
4Teams_Vacation4	sec			
8Teams_Vacation1	1.2 sec	1100.4	9.42%	12 min 49 sec
8Teams_Vacation2	sec			
8Teams_Vacation3	sec			
8Teams_Vacation4	sec			
16Teams_Vacation1	2.53 sec	2191.8	9.38%	21 min 31 sec
16Teams_Vacation2	sec			
16Teams_Vacation3	sec			
16Teams_Vacation4	sec			
32Teams_Vacation1	4.6 sec	4254.2	9.11%	63 min 19 sec

32Teams_Vacation2				
32Teams_Vacation3				
32Teams_Vacation4				

Vacation Scenarios 2, 3 and 4 to be run on reference machine for publication.